

Shell Scripting – Assessment Test

Total Marks: 50 | Time: 1 Hour

Section A – Short Answers (15 Marks)

1. What is a shell script? Why do we use it in Linux?
2. What is the use of `#!/bin/bash` in a script?
3. Explain the difference between environment variables and shell variables.
4. What does the command `systemctl is-active --quiet httpd` do?
5. What is the purpose of `set -e` and `set -x`?
6. What is the difference between `read`, `readline`, and `readlines` in Python?
7. What does the exit status (return code) represent in shell scripting?

Section B – Write the Output (10 Marks)

Q8.

```
x=5
if [ $x -gt 3 ]
then
    echo "greater"
else
    echo "less"
fi
```

Q9.

```
for user in dev test admin
do
    echo "User: $user"
done
```

Q10.

```
echo "Pratiksha" | cut -c 1-3
```

Q11.

```
echo -e "apple\napple\nbanana" | sort | uniq
```

Q12.

```
touch file1
```

```
if [ -f file1 ]; then echo "yes"; else echo "no"; fi
```

Section C – Concepts & Commands (15 Marks)

13. Explain the use of the following commands with examples: cut, sort, uniq, head, tail

14. Write a shell script to read a filename from the user and display the number of lines in that file.

15. Write a script to check if a directory exists. If not, create it.

16. What is heredoc? Write an example using: cat <<EOF

17. Write a script to backup a folder using tar with timestamp.

Bonus Question (Optional) (10 Marks)

Write a function in shell scripting that accepts a service name and prints whether it is active or not.